Rachamalla Preetham

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.96
Sri Chaitanya Junior Kalasala	Class XII (TSBIE)	2023	98.30%
Dr. KKR Gowtham High School	Class X (TS-SSC)	2021	100.00%

SKILLS

Languages: C, C++, Python, JavaScript, HTML/CSS, SQL

Frameworks: PyTorch, TensorFlow, Transformers, Matplotlib, ReactJS, scikit-learn, Seaborn, NumPy, Pandas, ChromaDB, LangChain, LangGraph, OpenCV

Technologies: Docker, Azure, GitHub/Git, MySQL, Java Spring Boot, PostgreSQL

Technical Skills: Full-stack web development, AI/ML solution/product development, Cloud Deployment

Soft Skills: Communication, Teamwork & Collaboration, Problem-Solving, Adaptability, Time Management

RELEVANT COURSES

Computer Science and Programming: Introduction to Programming, Data Structures and Algorithms, Algorithms, Introduction to Artificial Intelligence, Introduction to Computing, Digital Fabrication, Theory of Computation

AI : Supervised Learning - Regression and Classification - Coursera, Advanced Learning Algorithms - Coursera, Transformers and LLMs - Hugging Face, RAG, Agentic AI, Multi Agent Systems, MCP.

Mathematics: Calculus 1, Calculus 2, Elementary Linear Algebra, Differential Equations, Introduction to Probability, Probability and Random Variables, Introduction to Statistics, Maths for Physics, Concentration Inequalities

WORK EXPERIENCE

Kotak Mahindra Bank -- Software Engineering Intern | Jan 2026 - Jun 2026

- Built and deployed **Embark**, a new customer-data lookup service replacing a Power BI-based workflow.
- Developed the **React/Vite micro-frontend** and backend, integrating with **Cloud Bank, Module Federation & UCDS** for end-to-end customer data retrieval.
- Managed **CI/CD and DEV/UAT deployment**, working across frontend, backend and platform teams.
- Added **channel-level TIBCO metrics** for the Credit Card team, enabling usage analysis through Grafana.

PROJECTS

FIRE-SAR: Multi-Agent Wildfire Search, Survivor Tracking & Rescue Coordination

- Architected a decentralized coordination framework for heterogeneous UAV swarms under dynamic communication topologies and spreading fire constraints to search, track, and rescue lost survivors.
- Implemented distributed task allocation using Multi-Agent Reinforcement Learning alongside an optimization-based rescue resource dispatch model , outperforming greedy and nearest-feasible baselines.
- Integrated Signal Temporal Logic (STL) to formally verify mission safety specifications (separation thresholds, risk limits, and bounded revisit deadlines) within a real-time interactive simulation and telemetry dashboard.

Multimodal RAG Ingestion & Retrieval Pipeline

- Engineered an automated multimodal RAG pipeline using Unstructured.io layout-aware title chunking to parse complex PDFs containing text, structured HTML tables, and figures.
- Leveraged GPT-4o Vision to generate query-dense summaries for visual and tabular data while preserving raw

assets in metadata for grounded, cross-modal ChromaDB retrieval.

Algorithmic Trading & Market Matching Engine

- Engineered a dual-phase trading engine in C++ comprising an autonomous trading client and a central limit order book market simulator.
- Implemented price-time priority order matching, trade execution settlements, and automated position/PnL tracking under strict latency constraints.

Custom Byte-Pair Encoding (BPE) Tokenizer Engine

- Built a production-grade Byte Pair Encoding (BPE) tokenizer from scratch, implementing regex-based pre-tokenization, iterative vocabulary training, and byte-level fallbacks.
- Engineered subword merge lookups and decoding trees for arbitrary UTF-8 byte streams, handling special control tokens and out-of-distribution token boundary artifacts.
- Benchmarked vocabulary compression ratios and token density across multilingual and code datasets to analyze context-window utilization efficiency.

Autonomous Code Repair & Migration Agent

- Built a closed-loop code maintenance agent that isolates environments in Docker, parses test failure traces, and iteratively generates validated Git patches.
- Implemented a persistent vector memory layer using Qdrant to capture successful repair rules, enabling single-shot resolution of recurring dependency conflicts across repositories.

POSITIONS OF RESPONSIBILITY

Elan & NVision Coordinator

OCS Internship Coordinator

Sunshine UG Mentor, E-Cell Marketing Manager, Milan Events Coordinator, Shuffle Coordinator

EXTRACURRICULAR

Hobbies: Dance, Cycling, Running, Reading Self-Improvement Books, Watching movies and Series, Playing Video Games and Sports & **NSO Cricket**